

Design and Study of Waveguide using HFSS-High Frequency Structural Simulator

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Abstract—The waveguide is a hollow tube which is used for transmission of microwave range frequency electromagnetic waves. Waveguides are divided into different types depending on their shapes. Waveguide in form of a rectangle is termed a rectangular waveguide. Rectangular waveguides are the earliest mode of transmission lines used for compact systems like radars and inside equipment shelter. Rectangular Waveguides are used to transfer electromagnetic energy or power from one point in the space to another efficiently. They are used in many applications such as radars, isolators, attenuators and slotted lines. A rectangular waveguide supports TM and TE modes but not TEM waves because we cannot define a unique voltage since there is only one conductor in a rectangular waveguide. An air filled Rectangular waveguide is simulated using HFSS (High Frequency Structural Simulator) simulation software to obtain different parameters. The electric and magnetic field patterns have been analyzed. Characteristic profiles are analyzed and the performance of the waveguide is studied. Theoretical analyses have been studied and both the results have been compared and verified. In addition to that the cutoff wavelength has been determined and compared with the theoretical value and its importance has been described which can be helpful in fabricating the real world components.

Keywords— Rectangular waveguide; Propagation constant; Characteristic impedance; Boundary conditions; Wave ports; HFSS.

1. INTRODUCTION

I waveguides:

Waveguides are basically a device for transporting electromagnetic energy from one end region to another end region. Typically, waveguides are hollow metal tubes like rectangular or circular in cross section. They are capable of directing power precisely to where it is needed, it can handle large amounts of power and function as a high-pass filter. At frequencies higher than 3 GHz, the transmission of electromagnetic waves along transmission lines and cables becomes difficult mainly due to the losses that occur both in the solid dielectric needed to support the conductor and in the conductors themselves. A hollow metallic tube of uniform cross section for transmitting electromagnetic waves by successive reflections from the inner walls of the tube is called a waveguide. A metallic tube can be used to transmit electromagnetic waves at these frequencies. They are used in UHF and microwave regions as an alternative to transmission

lines. It may be noted that, the TEM wave cannot exist in a waveguide but TE and TM waves can exist.

II Rectangular Waveguides:

The electromagnetic wave inside a waveguide can have an infinite number of patterns which are called modes. We know that an electromagnetic wave consists of magnetic and electric fields which are always perpendicular to each other. The fields in the waveguide which make up these mode patterns must obey certain physical laws. At the surface of a conductor, the electric field cannot have a component parallel to the surface. This indicates that the electric field cannot have a component parallel to the surface. This indicates that the electric field must always be perpendicular to the surface at a conductor. The magnetic field on the other hand is always parallel to the surface of the conductor and cannot have a component perpendicular to it at the surface.

In general, there are two kinds of modes in a waveguide. In the first type, the electric field is always transverse to the direction of propagation and is called the **Transverse Electric** or **TE** wave. In the second type, the magnetic field is always transverse to the direction of propagation and is called the **Transverse Magnetic** or **TM** wave. . Thus in a TE mode, no electric line is in direction of propagation i.e., $E_z=0$, If z is the direction of the propagation. But H_z not equal to zero. In a TM mode, no magnetic line is in direction of propagation i.e., $H_z=0$ but E_z not equal to zero.

We use subscript for designating a particular mode. TE_{mn} or TM_{mn} where 'm' indicates the number of half wave variations of the electric field (or magnetic field in a TM) across the wider dimension a , of the waveguide and 'n' indicates the number across the narrow dimension. Referring to TE pattern in Figure it can be seen that the voltage varies from '0' to maximum and maximum to '0' across the wide dimension a . This is one half variation. Hence $m=1$. Across the narrow dimension, there is no variation in voltage v . Hence $n=0$. Therefore this mode is TE_1 mode. The mode having the highest cut off wavelength is known as Dominant mode of the waveguide and all other modes are called higher modes. For example TE_{10} is the dominant mode for TE waves. It is the mode which is used for practically all electromagnetic transmission in a rectangular waveguide. Dominant mode is almost always a low-loss, distortion less transmission and

higher modes result in a significant loss of power and also undesirable harmonic distortion.

2. PROPOSED METHOD

In our project we are using simulation software in order to simulate the working and to find the characteristics of a waveguide. Presently in order to develop a waveguide we use a normal theoretical approach, where as in this project we get to see how the waveguide behaves for different set of operating frequencies.

Here we first design a waveguide of specific dimensions which we want for our application. For that we draw it in the HFSS software. Then we can select the type of material with which the waveguide is to be made. We can also select the hollow space in the waveguide to be either air or vacuum.

By use of this method we can find all the characteristics of the waveguide such as characteristic impedance, propagation constant, different modes of operation, boundary conditions and propagation of different fields without physically implementing the waveguide. This method is very much useful than the normal approaches of designing a waveguide.

3. DESIGN IMPLEMENTATION OF HFSS

HFSS is a commercial finite element method solver for electromagnetic structures from Ansys. The acronym originally stood for HIGH FREQUENCY STRUCTURAL SIMULATOR. It is one of several commercial tools used for antenna design, and the design of complex RF electronic circuit elements including filters, transmission lines, and packaging [12].

HFSS is a high-performance full-wave electromagnetic (EM) field simulator for arbitrary 3D volumetric passive device modeling that takes advantage of the familiar Microsoft Windows graphical user interface. It integrates simulation, visualization, solid modeling, and automation in an easy-to-learn environment where solutions to your 3D EM problems are quickly and accurately obtained. Ansoft HFSS employs the Finite Element Method (FEM), adaptive meshing, and brilliant graphics to give you unparalleled performance and insight to all of your 3D EM problems. Ansoft HFSS can be used to calculate parameters such as S-Parameters, Resonant Frequency, and Fields.

The name HFSS stands for High Frequency Structure Simulator. Ansoft pioneered the use of the Finite Element Method (FEM) for EM simulation by developing/implementing technologies such as tangential vector finite elements, adaptive meshing, and Adaptive Lanczos-Pade Sweep (ALPS). Today, HFSS continues to lead the industry with innovations such as Modes-to-Nodes and Full-Wave Spice [12].

Typical uses include:

- Package Modeling – BGA, QFP, Flip-Chip

- PCB Board Modeling – Power/Ground planes, Mesh Grid Grounds, Backplanes
- Silicon/ GaAs - Spiral Inductors, Transformers
- EMC/EMI – Shield Enclosures, Coupling, Near-or Far-Field Radiation
- Antennas/Mobile Communications – Patches, Dipoles, Horns, Conformal Cell Phone Antennas, Quadrafilair Helix, Specific Absorption Rate(SAR), Infinite Arrays, Radar Cross Section(RCS), Frequency Selective Surfaces(FSS)
- Connectors – Coax, SFP/XFP, Backplane, Transitions
- Waveguide – Filters, Resonators, Transitions, Couplers
- Filters – Cavity Filters, Microstrip, Dielectric.

Ansoft HFSS has evolved over a period of years with input from many users and industries. In industry, Ansoft HFSS is the tool of choice for high-productivity research, development, and virtual prototyping.

4. DESIGN AND ANALYSIS OF WAVEGUIDE USING HFSS

Creating designs, involves the following:

- 1. Parametric Model Generation** – creating the geometry, boundaries and excitations
- 2. Analysis Setup** – defining solution setup and frequency sweeps
- 3. Results** – creating 2D reports and field plots
- 4. Solve Loop** - the solution process is fully automated.

The Process

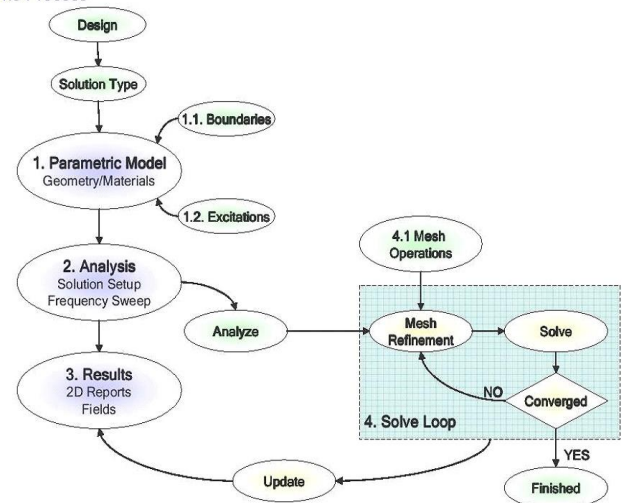


Fig. 1. The Process of steps involved in HFSS for designing the waveguide.

The following steps are to be followed in order to design the waveguide:

Step I: On the **Project** menu, click **Insert HFSS Design**

Step II: On the **HFSS** menu, click **Solution Type**. The **Solution Type** dialog box appears.
Select Driven Modal in the solution types.

Step III: Setting the Model's Units of Measurement.

Step IV: Drawing a Model.

Step V: Assigning Materials.

Step VI: Assigning Boundaries.

Step VII: Assigning Excitations.

Step VIII: Solution Setup.

Step IX: Frequency Sweep setup.

Step X: Running a Simulation.

Once the simulation is completed HFSS informs you in the message window.

5. RESULTS

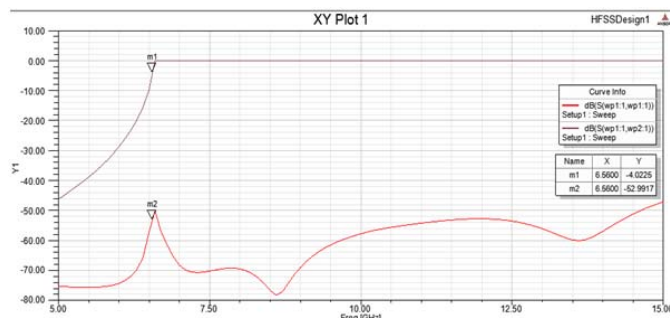


Fig. 2. Propagation constant

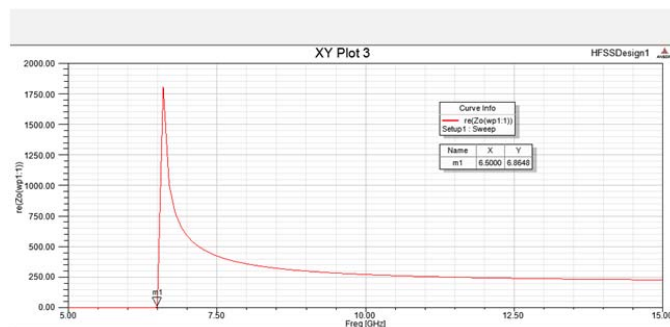


Fig. 3. Characteristic impedance

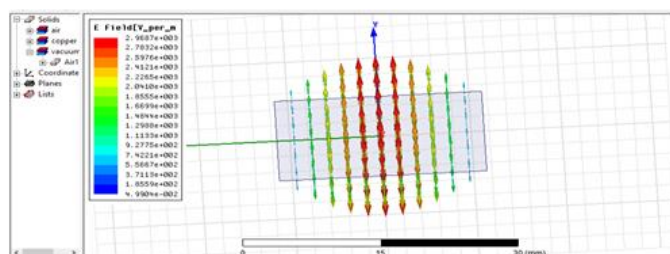


Fig. 4. Electric Field Pattern

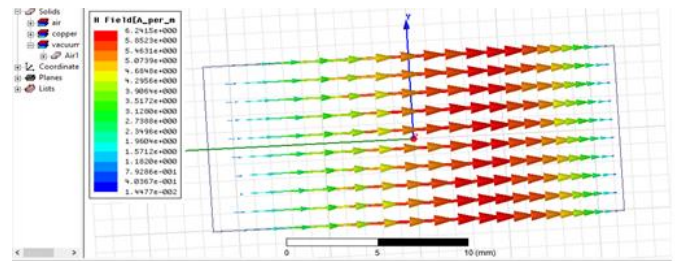


Fig. 5. Magnetic Field Pattern

Above figure(2) shows that the waveform of the propagation constant of designed waveguide, figure(3) shows that waveform of the characteristic impedance of the designed waveguide, figure(4) shows that electric field pattern of the designed waveguide, figure(5) shows that the magnetic field pattern of the designed waveguide.

6. CONCLUSION

In this project, we have designed waveguides using HFSS (High frequency structural simulator). It has been observed from the results that the designing characteristics of the waveguides can be analyzed, implemented and designed quite better than the normal approaches of designing a waveguide.

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